

Design and Implementation of a Multi-Level Dynamic Key Management Scheme for Secure Aeronautical Broadband Communication Systems

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Abstract

In order to meet the high-speed, low latency and high reliability requirements of aviation broadband communication scenarios, this study proposes a light multi-level dynamic key management solution for safe aviation broadband communication systems, which is designed to include root keys, main keys and session density secrets. The three-layer key structure of the key to improve safety and efficiency. The national secret algorithm provided by the domestic password card hardware has achieved symmetrical key generation, distribution, dynamic updates and security storage mechanisms. The method proposed by this study can ensure the security of the forefront, minimize risks and reduce key management overhead, which provides a key guarantee for the data security transmission process of the aviation broadband communication system.

CCS Concepts

• **Security and privacy** → Cryptography; Key management.

Keywords

Aeronautical Broadband Communication System, Multi-level Keys, Dynamic Key Update, Key Management, Key Derivation

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1 INTRODUCTION

With the continuous progress of aviation broadband communication technology, the demand for security data transmission is becoming more urgent [1]. Such systems have a key role in ensuring real-time data interaction between aircraft and ground stations. However, the high dynamic and low-delay requirements of the

aviation environment have brought severe challenges to the traditional key management mechanism. It is difficult to fully meet the special needs of the aviation environment, especially in key dynamic updates and defense network attacks.

As the core of communication security, key management is crucial to the confidentiality, integrity and authenticity of the data [2] [3]. When processing key generation, allocation, and update, traditional methods are often accompanied by higher computing expenses and latency [4] [5], which is particularly unfavorable for aviation systems that require low latency and high-efficiency data transmission. In addition, as users and data scale expand, the risk of key leakage has increased significantly. Therefore, the development of a more flexible and lightweight key management solution is particularly urgent.

Both the industry and academic circles have conducted in-depth research on key management issues. Aeronautical Radio Inc. (ARINC) proposed two types of security talks based on public key/private key [4] and pre-shared key [6]. However, these protocols have problems such as insufficient password complexity, and susceptible to key leak attacks [7] and limited encryption kit. In addition, the progress of the L-band Digital Aviation Communication System (LDACS) introduced a stronger security mechanism, such as end-to-end encryption and certification protocol [8] [9], while NIST's key derivation function [10] [11] then then Further regulates the practice of key management, make up for the existing defects, and enhances the robustness of the key management process.

This study proposes a lightweight multi-level dynamic key management solution to deal with the above challenges. This scheme adopts hierarchical key structure, including three layers of root keys, main keys and session keys to achieve efficient management and enhanced security. By adopting China's national standard encryption algorithm, high-speed key operations are guaranteed, but also meet national standards. The core functions of the plan include safety key generation, dynamic key distribution, high-efficiency key renewal and reliable key storage.

The main contributions of this study include:

1. The multi-level key management framework that adapts to the different security levels and dynamic environments that adapt to the aviation broadband communication system.
2. A high-efficiency key generation mechanism is proposed, which significantly reduces delay and calculation costs.
3. Realize the security key distribution and update strategies, effectively prevent key leakage, and ensure the security during the switching process.

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The structure arrangement of the article is as follows: The second part explains the implementation method of the plan mentioned in detail; the third part introduces the architecture and specific implementation of the key management system; the fourth part discusses the experimental results and security assessment; And look forward to the future research direction.

2 METHODOLOGY

In this study, a set of light multi-level dynamic key management solutions are designed for the aviation broadband communication system, which aims to meet the challenges such as dynamic scenes, low latency requirements and high security standards. This scheme combines the storage mechanism of layered key structure, efficient key generation, security distribution, dynamic renewal and reliable storage mechanism. The following sections will overlook the core component of the proposal.

2.1 Multi-Level Key Structure Design

The proposed approach adopts a hierarchical key structure to enhance security, minimize risks, and ensure efficient management. The structure consists of four key types:

Root Key (KAS) is used for identity authentication as the basis of the main key negotiation between the aircraft system (AS) and the security gateway (SGW). This key is manually distributed manually before the system initialization. Master Key (KAS-SGW, KAS-GS): is sent by the root secret to share between AS and SGW or ground station (GS) to generate session keys to ensure safe communication between system components. Session Key (KU-ENC, KU-INT), Dynamically generated for encrypting user data and ensuring data integrity. These keys are specific to individual communication sessions and are updated regularly. Key Encryption Key (KEK) are used to encrypt and protect stored keys. This key ensures the separation and confidentiality of all other keys in the system.

Each layer of keys is designed for specific functions, providing forward security and isolating risks. Even if the low-level key is damaged, the high-level keys are still safe, thereby the general affected by the threat to minimizing the system.

2.2 Key Generation Mechanism Design

The key generation process is critical to ensure randomness and security, which uses the following two approaches.

Hardware-Based Random Number Generation: Root keys and KEKs are generated using physical noise sources in the cryptographic hardware card, ensuring high-quality randomness.

Key Derivation Function (KDF): Master and session keys are derived using the Password-Based Key Derivation Function (PBKDF)^[12], compliant with GM/T 0091-2012 standard. The KDF takes the root key, a random salt, and iteration counts as inputs to generate derived keys. This method ensures high entropy and meets the specific requirements of the aeronautical communication system.

The PBKDF process is illustrated In Figure 1, which highlights the use of cryptographic algorithms such as SM3-HMAC for enhanced security and reliability.

Master and session keys are derived using the PBKDF algorithm based on shared root keys, random salts, and system parameters.

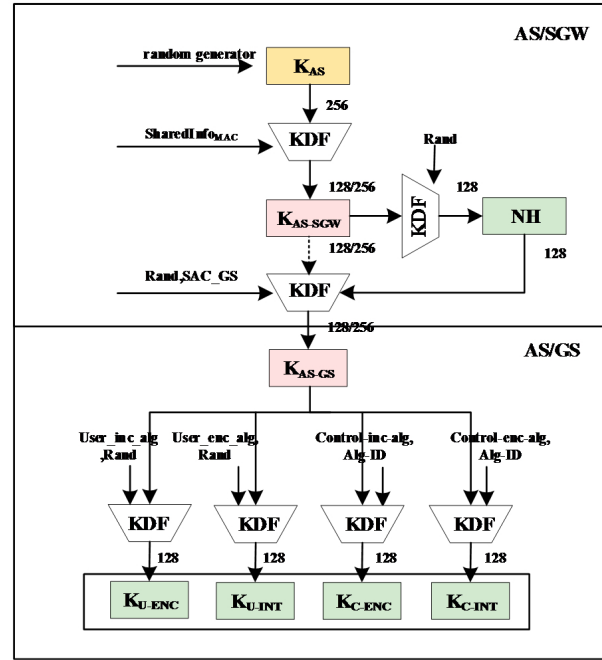


Figure 1: PBKDF Process for Key Derivation.

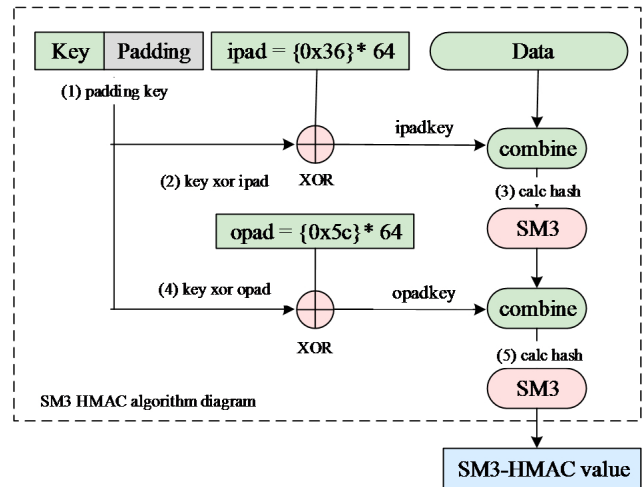


Figure 2: SM3 HMAC algorithm diagram.

As shown in Figure 2, SM3-HMAC is used as the cryptographic hash function to ensure strong entropy and security.

2.3 Key Distribution Protocol Design

Secure key distribution is a critical component of the proposed scheme, which ensures that keys are transmitted securely between AS, SGW, and GS:

Root key is pre-shared and stored securely in the cryptographic hardware of AS and SGW. Master key is generated during secure

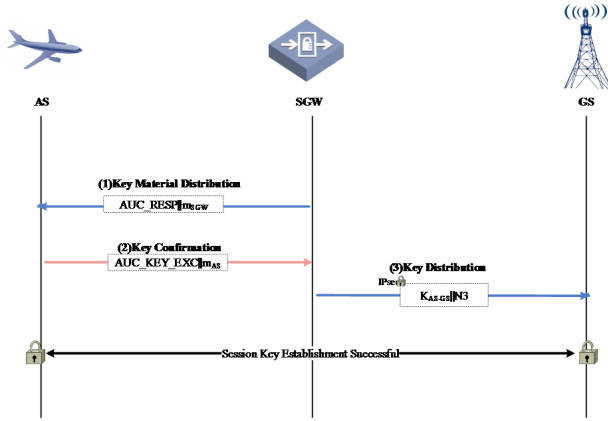


Figure 3: Secure Key Distribution Flow.

access authentication and distributed from SGW to GS using an IPSec-secured channel. Session keys are derived locally by AS and GS using shared master keys and random numbers. This approach reduces the need for frequent transmission of sensitive keys to enhance security.

A detailed flow of the key distribution process between AS, SGW and GS is shown in Figure 3.

The process entails three stages: key distribution from SGW to AS, AS key confirmation, and key distribution from SGW to GS. Initially, SGW generates K_{AS-SGW} using KDF, encrypts an AUC_RESP message, and sends it to AS. AS verifies, derives K_{AS-GS} and session keys, and encrypts a confirmation back to SGW. SGW validates, forwards K_{AS-GS} to GS, who then derives session keys, completing the AS-GS key exchange.

2.4 Dynamic Key Update Mechanism Design

In order to reduce security threats such as password cracking and leakage, the keys need to be updated regularly in the aviation broadband communication system. The root keys are manually updated once a year, and the distribution process is re-executed. The master key K_{AS-SGW} is re-generated when the aircraft and the new gateway are accessed. The master key K_{AS-GS} is dynamically updated during GS switching on the ground station, while updating its derived session key.

Using the K_{AS-SGW} key for data encryption entails a risk, as any leakage of the root secret key could compromise the security of all encrypted communications. In order to improve security, the key generated by negotiated is instead of the K_{AS-SGW} key. When GS switching, the main key can be replaced without re-authentication to increase the flexibility of security protection and the security of communication. The key update process is illustrated as Figure 4.

During base station handover, the target base station alerts the security gateway of a key update, including relevant identifiers. The gateway requests an update from the AS, which responds with updated keys via KDF. The SGW synchronizes and computes a new K_{AS-GS} , then sends key types, main keys, and random numbers to the target GS. The GS updates its session key accordingly.

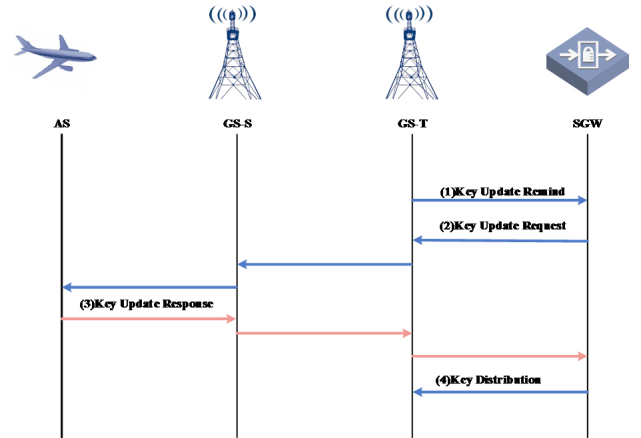


Figure 4: Key Update Process in Dynamic Scenarios.

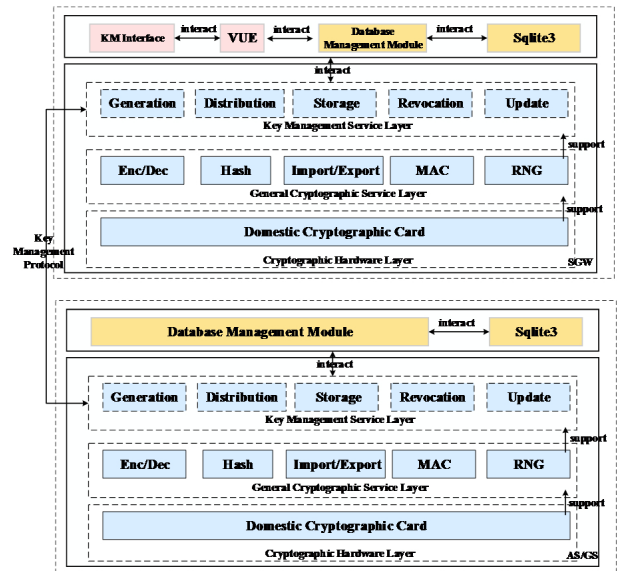


Figure 5: System Architecture Overview.

2.5 Key Storage and Revocation Design

In order to protect the integrity of the key and prevent unauthorized access, the plan uses the following measures. All keys use KEK encryption and store in the safety hardware encryption module. KEK encrypts keys in a one-to-one manner, ensuring the separation of keys. In the event of key revocation, its status in the system database is marked, and its child keys are also revoked.

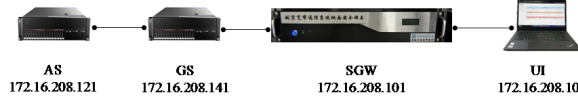
2.6 System Architecture Design

AS shown in Figure 5, The key management system consists of the gateway end, the AS end and the GS.

The application layer of the key management system is a cryptographic hardware device layer, a general-purpose password service layer and a key management service layer from the bottom of the

Table 1: Implementation Environment of the System.

Item	Configuration/description
Gateway Device	Industrial control machines with PCI-E cryptographic hardware cards
CPU	8-core processor.
RAM	8GB DDR4.
Storage	512GB SSD.
AS and GS Cryptographic Support	Mini PCI-E cryptographic cards
Operating System	Ubuntu Linux.
Development Tools	Back-end: Go and C programming languages; Front-end: VUE3.
Database	SQLite3 for key storage and management.

**Figure 6: Physical connection deployment diagram of the communication testing platform.**

bottom. The password card is the hardware support of the password service layer. The universal password service layer shields the hardware details of the password device for the key management service layer, and the key management service layer provides key management functions. The database management system is implemented by SQLITE3. The basic operation of the data is completed through the database management module interact with the application layer.

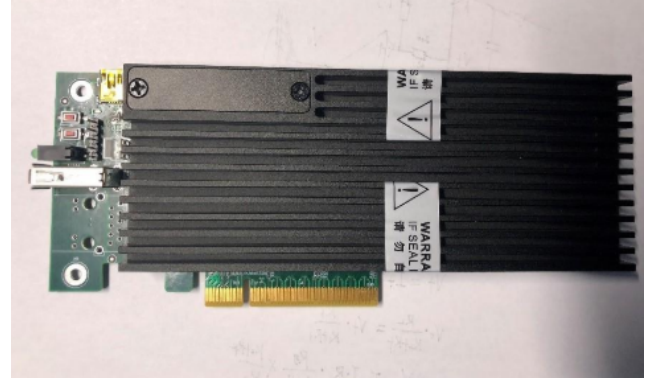
2.7 Development Hardware Design

As illustrated in Table 1 Implementation Environment of the System. The development environment includes both hardware and software components to support the implementation of the proposed scheme.

AS, GS and SGW are equipped with PCI-E cryptographic cards which supports SM2/SM3/SM4 algorithms, high-performance encryption and decryption, random number generation, and key storage function. During the implementation process, three servers equipped with cryptographic card hardware were utilized as AS, GS and SGW respectively. The semi-physical platform topology, the design of the Cryptographic hardware is shown in Figure 6 and Figure 7.

3 RESULTS ANALYSIS AND DISCUSSION

This section introduces the experimental results of the multi-level dynamic key management scheme, and the analysis results are analyzed to evaluate its performance, efficiency and adaptability under the aviation broadband communication scheme.

**Figure 7: cryptographic hardware card****Table 2: Key management efficiency.**

Method	Time consumed
Key Generation	0.195 ms
Root key distribution	No runtime latency
Root Key File Generation	1.123 ms
Root Key File Export	0.487 ms
Root Key File Import	6.241 ms
Key Import	0.830 ms
Key Handle Acquisition	3.609 ms
Key Derivation	222 ms
Key Storage	4.181 ms
Key Revocation	25 ms

3.1 Key Management Performance

As shown in Table 2, the measurement shows that the average root-in key is consumed by 0.195 ms. The root keys are pre-stored, thus it does not affect the operation time of the system. the time for derive key is 222ms, session keys were derived locally at AS and GS, eliminating the need for transmission. The time for the key to revoke is 25ms. The key derivative and local operations do not affect the time of the system message transmission, which is enough to meet the low delay requirements of the aviation system.

3.2 Security Evaluation

3.2.1 Resistance to Attacks. The proposed scheme was tested against potential security threats, including brute-force attacks and key compromise.

Brute-Force Resistance: The key length used in this study is at least 128 bits. Moreover, the use of high-entropy keys and SM3-HMAC for key derivation ensured resistance to brute-force attacks, requiring 2^{128} operations to break.

Key Compromise Mitigation: The hierarchical key structure minimized the impact of key compromise. For example, if a session key was compromised, the damage was limited to the affected session, and other keys remained secure.

3.2.2 Forward Secrecy. The key management system introduced in this research adopts a forward-secure key derivation mechanism,

which rigorously conforms to the PBKDF2 (Password-Based Key Derivation Function 2) standard. The generation process of new keys ensures their non-derivability from previous keys via reverse engineering, thereby effectively safeguarding the forward security of the system.

In the context of key updating, we have implemented a multi-level key hierarchy to securely derive the master key K_{AS-GS} between the AS and GS. This derivation process is based on the master key shared between the AS and SGW, ensuring both security and traceability during the key update process. More crucially, even if the key between the AS and GS is unfortunately compromised, the presence of the multi-level key hierarchy poses no potential threat to the master key K_{AS-SGW} shared between the AS and SGW.

3.2.3 Key Storage security. All keys were securely stored using key encryption keys within qualified crypto HSM. Integrity verification with hash-based message authentication codes (HMAC) detected no unauthorized modifications. The crypto hardware card complies with national standards and specifications such as GM/T 0018 "Specification for cryptographic Device Application Interface" and GM/T 0028 "Security Technical Requirements for Cryptographic Modules."

4 CONCLUSION

This study has designed a multi-level dynamic key management solution for aviation broadband communication data chains, including multi-level key structures and key distribution, storage, key dynamic update strategies and revocation methods. The aviation broadband communication data chain multi-level dynamic key management scheme involves the process management of the key life cycle of the key, and uses the use of dual-path physical noise sources as the source of the key to ensure the randomness and unpredictability of the key to ensure the key. The security of the key storage and usage by using a second-level PCIE password card storage key. The symmetrical key management solution designed by this research group provides a solid key guarantee for the data security transmission process of the aviation broadband communication system. During the process of key management, the use of high-speed national secret algorithm and lightweight symmetrical

key management process is suitable for the use of aviation broadband communication systems with high speed, low latency special points. The development and deployment laid a solid foundation.

Acknowledgments

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